EUROPEAN SYNCHROTRON RADIATION FACILITY
INSTALLATION EUROPÉENNE DE RAYONNEMENT SYNCHROTRON

ESRF

Standard Application for Beam Time at the ESRF

Proposal Summary

Title

Arrested spinodal demixing of aqueous solutions of small inorganic solutes

Keywords

#1: phase separation #2: nonclassical nucleation/ crystallization #3: liquid-condensed precursor #4: amorphous

Abstract

Arrested spinodal demixing (SD) provides nanoporous materials with high surface area, excellent pore accessibility, and structural isotropy. These traits are sought-after for sensing, catalysis, or energy applications. However, SD is still deemed inapplicable for aqueous solutions of small inorganic solids. Tenets claim that the instability regime is inaccessible as burst nucleation intervenes. With our approach combining unprecedented fast-flow chemistry with spontaneous formation of pre-critical coordination entities (eg, prenucleation clusters), SD becomes accessible in a wide range of inorganic solutes, ranging from mineral-forming main group compounds to transition-metal MOF-type systems. In this experiment, in-situ small- to ultrasmall-angle X-ray scattering will provide the missing piece of evidence for genuine spinodal demixing via the occurence and temporal evolution of a dominant spinodal length and its centrosymmetric scattering patterns.

- **This proposal is:**
  A new proposal ☑
  A resubmission of ... ☐
  A continuation of : ... ☐
- This proposal is:
  Fundamental Science ☑ % 100 Applied Science ☐ % 0 Industrial Science ☐ % 0

Societal Themes

- ○ Earth and Environment
- ○ Energy
- ○ Health
- ◉ Fundamental Science
- ○ Information and Communication Technology (ICT)
- ○ Other
- ○ Other Functional Materials

Scientific Area of the proposal

- ○ HC - Hard Condensed Matter Science
- ○ SC - Soft Condensed Matter Science
- ○ ES - Earth Sciences
- ○ MA - Applied Materials Science
- ○ LS - Life Sciences
- ○ EV - Environment
- ○ ME - Engineering
- ○ MD - Medicine
- ○ HG - Cultural Heritage
- ◉ CH - Chemistry
- ○ MI - Methods and Instrumentation

Main proposer (to whom correspondence will be addressed):

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Co-Proposers (Laboratory if different from main proposer)

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Beamline(s) requested:

Principal # 1: ID02
OR
(alternatives)
...
...

Total required shifts: 6

Société Civile
RCS Grenoble D 338 723 919
N° Siret 338 723 919 00027 - APE 7219 Z

Postal Address: ESRF User Office - CS 40220 - 38043 Grenoble Cedex 9 - France
Location: Polygone Scientifique Louis Néel - 71, avenue des Martyrs - 38000 Grenoble - France
Telephone (Switchboard): 33-(0)4 76 88 20 00 - Telefax: 33-(0)4 76 88 20 20

Number of shifts required 6

Preferred starting time: *Please select the period* December/January *Unacceptable dates* 30/06 to 30/09 (parental leave of SEW)
23.10.2023 - 04.11.2023
26.08.2023 - 03.09.2023

Beam Requirements

☐ 16 Bunch Mode ☐ 4 x 40mA Mode ☑ Multi Bunch
☐ Circular polarization ☐ White beam ☐ Monochromatic beam
☑ Fixed energy [keV]: 12.0 ☐ Tunable energy [keV] from: ☐ to: ☐
Beam energy resolution [meV]: Spot size on sample [µm]:
Other:

Laboratory Support Facility

- ☐ BioMedical Facility
- ☑ Central Chemistry Lab (Science Building)
- ☐ Electrochemistry Lab LOB-023
- ☐ NFFA Access
- ☐ PSCM Labs (Science Building)
- ☐ Protein Support labs

Sample Environment

Items Supplied by the ESRF

☐ Furnace ☐ Magnet ☐ Cryostat ☐ Cryogenic gas stream ☐ Refrigerator
☐ Laser Class ... Wavelength [nm]
☐ High pressure Pressure range [GPa] from to
☐ Fixed temperature Temperature range [K] from 273.0 to 303.0
Temperature [K]
Detector system
Other equipment SFM-3000 or SFM-4000 Stopped-flow (Biologic), ideally thermosated.

Items Not Supplied by the ESRF

List all equipment that you will insert into the instrument

☐ Laser Class Wavelength [nm]
☐ Other equipment
Please indicate requirements for special equipment or facilities

Sample Description

Substance and formula

Calcium carbonate CaCO3
Cobalt malonate Ca(C3H2O4)
Cobalt hydroxycarbonate Co2CO3(OH)2

☐ Single crystal ☑ Powder ☑ Polycrystalline ☐ Multilayer ☐ Liquid ☐ Gas

☐ Nanoparticles ☐ Prepared at ESRF

☐ Other in situ generated precipitate

Average size [mm] 0.001 Volume [mm³] 1000.0 Surface area [mm²]
Mass [mg] Matrix or solvent water Conc.of absorb.[mmol]
Space group Cell dimensions at T= K:
a= Å b= Å c= Å alpha= Â° beta= Â° gamma= Â°

Container (capillary, flat plate, type of pressure cell, etc.)
Extra information required for Macromolecular Crystallography:
Origin and expression system
Previously observed diffraction (resolution, X-ray source, exposure/Â°)

Safety

Is the sample:

☐ Radioactive? ☐ Contaminant? ☐ Corrosive? ☐ Oxidizing?
☐ Explosive? ☐ Biologic? ☑ None of those

Is there any danger associated with the proposed sample, with any preparation at ESRF, or with sample equipment?

○ Yes ○ Uncertain ◉ No

If you have ticked Yes or Uncertain, you must give details of the associated risks in the box below:

Will you use live animals *on site for your experiment* ***(all kinds of animals are concerned)?*** ○ Yes ◉ No

After the experiment *, will the sample be:* ◉ Removed by user? ○ Stored at ESRF?

To be filled by ESRF ***Sample environment code:***	***Comments by safety Officer:***

Experience with Synchrotron Radiation

What are the technical reasons which make ESRF necessary for your experiment?Why are other synchrotron radiation sources not appropriate?

For detection of spinodal decomposition, we need high brilliance for appropriate time resolution, USAXS for appropriate monitoring of the evolving spinodal length, SAXS for probing of the initial formation of the nanometers prenucleation species. The combination of these requirements points directly at ID02 at ESRF; besides the technical demands to the beamline, it is especially advantageous that an identical stopped-flow device is already in use as a standard sample environment at ID02, saving us and the beamline from extensive alteration/setup for accommodating an stopped-flow device to the beamline.

Have you used synchrotron radiation at the ESRF?	○ No	◉ Yes	
Have you used synchrotron radiation at other sources?	○ No	◉ Yes, at:	BESSY, Diamond, DESY, ALBA
Have you already used synchrotron radiation for this project?	○ No	◉ Yes	

Publications

Please give the references of papers published by the proposers during the past 3 years as a result of experiments done at the ESRF.

Origin (1): if from data from ESRF beamlines ONLY, (2) : if from data from more than one source

	Description	Origin
[1]	Template-free synthesis of mesoporous and amorphous transition metal phosphate materials | Karafiludis S., Buzanich A., Heinekamp C., Zimathies A., Smales G.J., Hodoroaba V., ten Elshof J.E., Emmerling F., Stawski T.M. | Nanoscale, p.epub, 2023	(2)
[2]	Nucleation pathway of calcium sulfate hemihydrate (bassanite) from solution: Implications for calcium sulfates on Mars | Stawski T.M., Besselink R., Chatzipanagisad K., Hovelmann J., Benning L.G., Van Driessche A.E.S. | Journal of Physical Chemistry C, vol.124, p.8411-8422, 2020	(2)

New ESRF User

New ESRF users may list below up to 10 publications not involving any data taken at ESRF. (Do NOT list any ESRF publications here, this MUST be done in the section above).

	Description	
[1]		

EUROPEAN SYNCHROTRON RADIATION FACILITY

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Application for beam time at ESRF – Project Description

Proposal Summary:

Although highly desired, Spinodal Demixing (**SD**) is not yet established for aqueous solutions of inorganic compounds. We developed a fast-flow approach to force SD in aqueous systems that form prenucleation species (**PNS**) by turbulent Ball-Berger mixing as provided by commercially available stopped-flow mixers (**SFM**), yielding bicontinuous, interpenetrating networks indicative of arrested SD. This experiment aims to use in situ and time-resolved USAXS at ID02 in combination with a BioLogic™ (already established as sample environment at ID02) to gather final experimental evidence for genuine SD for three model systems: calcium carbonate ($CaCO_3$·n H_2O), cobalt hydroxy carbonate ($CoCO_3$·$[Co(OH)_2]_n$·n H_2O), and cobalt malonate ($[CoCH_2(COO)_2$·n H_2O]-malonate). The systems differ in separation kinetics and structural features of the arrested spinodal networks, allowing us to establish arrested SD as a phenomenon generally accessible in PNS-forming systems. We will follow the structure formation and evolution by monitoring the dominant spinodal length (~ 100-500 nm) and its temporal evolution from tens of milliseconds to minutes.

Scientific background:

Arrested SD yields bicontinuous, isotropic nanonetworks with excellent pore accessibility and high surface area, decisive traits for sensing, sorption, catalysis, or energy applications. Although a fundamental phase separation mechanism, SD is still limited to alloys, glasses, and polymers. SD is categorically dismissed as infeasible for small inorganic solutes because SD is experimentally challenging: the system has to cross the metastable zone without inducing nucleation. It is deemed infeasible due to burst crystal nucleation, especially in water, preventing synthesis of nanonetworks "simply by mixing". But this view ignores PNS formation that counteracts crystal nucleation [1], *e.g.*, nanometer-sized prenucleation clusters of $CaCO_3$ [2-4]. In many solute systems, these coordination entities form before phase separation, promoting solid-amorphous phases. We recently showed that amorphous $CaCO_3$ (**ACC**) forms from aggregating, nucleating PNS [3, 4]. Our novel fast-flow approach enforces SD in PNS-forming systems, using turbulent Ball-Berger mixing in a standard BioLogic™ SFM. It yields amorphous bicontinuous nanonetworks whose structural features indicate SD, see Fig. 1B-D. It is broadly applicable, e.g., for mineral-forming compounds (e.g., ACC, Fig. 1B), transition metal compounds (e.g., $CoCO_3$·$[Co(OH)_2]_n$·n H_2O, Fig. 1C) or MOF-analogues compounds (Co-malonate, Fig. 1D). The characteristic bicontinuous networks are usually preserved due to a structural arrest which is sensitive to reaction conditions. The arrest can be delayed so that the initial network collapses into liquid-condensed droplets (see Fig. 1B* for the case of ACC). Cryo-TEM snapshots of earlier stages show that these droplets originate from nanonetworks (similar to Fig. 1B).

We have extensively characterized our approach. SANS with contrast variation showed an extremely hydrated inorganic network for the arresting calcium carbonate system, which is structurally stable for hours.

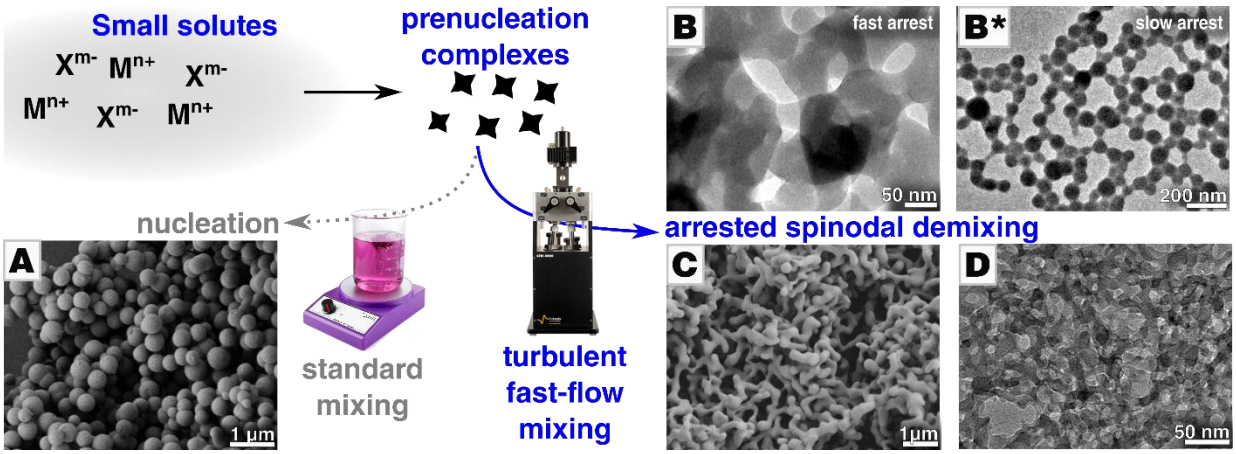


Figure 1. Nucleating prenucleation complexes form amorphous particles (**A,** amorphous $CaCO_3$); arrested SD yields nanonetworks, e.g., **(B)** amorphous $CaCO_3$, **(C)** basic $CoCO_3$, **(D)** cobalt malonate. **(B*)** shows spinodal ACC under slow arrest conditions, the network (B) collapses into droplets (B*).

Turbidometry —by SFM-coupled UV/Vis— evidenced instantaneous structure formation for all network-forming systems within few milliseconds (indicative for SD), while non-fast flow gave standard nucleation processes with induction times of seconds. X-ray and cryo-electron tomography verified our networks to be bicontinuous, interpenetrating, and structural isotropic— further evidence for SD. Still, we lack final proof for SD. The ultimate SD criterion is "continuous but rapid growth of sinusoidal composition modulations of a certain wavelength" [5]. Scattering is the gold standard to verify SD; it reveals SD via a broad correlation peak whose position is proportional to the size of the scattering features, gradually shifting position with the temporal evolution of SD [5]. The isotropic scattering patterns can be interpreted through well-established mathematical models [5,6]. Based on our characterization (e.g., Fig. 1), our nanonetworks are bicontinuous systems with a correlation length of up to several hundred nanometres, ideally suited for USAXS at ID02.

Experimental plan:

All precipitations are performed as double decomposition reactions; thus, two reactant solutions with suitable solution concentrations and composition (pH) will be mixed in the SFM. The device carries an observation head comprising a borosilicate capillary (ID 1.5 mm, opening of 60°) that will be aligned with an X-ray beam (Fig. 1). Each experimental run will consist of a flow phase and a subsequent stop phase (zero flow, sample ages in the capillary). Continuous monitoring across both phases allows probing different reaction stages in the continuous flow stage, optimizing the signal-noise ratio due to the extended acquisition period. The stop phase details the networks' structural evolution, profiting from ID02's high detector time resolution. All of the below reaction sets have already been cross-checked and characterized by our earlier experiments. Three model systems will be examined: (a) calcium carbonate, (b) cobalt hydroxy carbonate, and (c) cobalt malonate. The following measurements will be carried out at USAXS configuration: Three standard reactant concentrations will be measured (10, 20, and 40 mM — supersaturation impacts, i.a., the networks porosity) plus one concentration below the spinodal and above the nucleation threshold serves as a control experiment for standard nucleation behavior. The high concentration runs (40 mM) will be measured with an extended stop phase to document late Ostwald ripening. The high and the low reaction concentrations will be measured at three additional different flow rates (1 ml s^{-1} to 15 mL s^{-1}). The modulation of the arrest behavior will be measured by three additional runs at 40 mM reactant concentration but appropriately adjusted solution composition. We will also perform measurements under SAXS configuration at non-saturated conditions, thus, of the homogenous solution state stable against SD/nucleation to trace the formation of nanometer-sized PNS as precursors to SD. We will further acquire all the necessary background and blank measurements for each system individually.

Beamline(s) and beamtime requested with justification:

The experiments will be performed at ID02. Our experiment will have to probe length scales from 50 to over 500 nm, corresponding to q_{peak} up to 0.01 nm^{-1}. We expect this q-value as the correlation peak maximum, hence $q_{min} < q_{peak}$; thus, the instrumental range of ID02 in USAXS configuration of $q > 0.001$ nm^{-1} fits perfectly (already consulted with the beamline scientist). Since our SD is on the time scale of ~10 ms, the available Dectris Eiger2-4M USAXS detector (~1000 fps) is ideal for probing the fast kinetics of these SD processes. The auxiliary Rayonix LX170 WAXS detector will monitor crystallinity in post-spinodal Ostwald ripening. The isotropic 2D patterns will be reduced to 1D, corrected for transmission and background, and the structural information will be extracted using analytical models for scattering for SD morphologies [5]. We use a standard BioLogic SFM already established at ID02 as a sample environment. Based on our experiment plan, we **request 6 shifts (2 days)** of beamtime: W/SAXS feasibility studies at the BESSY µSpot taught us that a triplicate of one experimental run plus thoroughly cleaning/refilling of the SFM requires 40 min on average. We measure 16 individual reaction/mixing conditions for each of the three systems (4 concentrations + three additional flow rates at high and low concentrations + 3 modulated arrest conditions + one extended stop phase = 16). Thus, we require 4 shifts for USAXS: 3 systems × 16 conditions × 40 min = 32 h. We assign 1 further shift for SAXS configuration (PNS probing) and 1 shift for sample setup, optimizing acquisition parameters acquisition of all blanks and background measurements.

Results expected and impact - analysis strategy and significance of the results:

This experiment is key to the DFG-funded Heisenberg project of S. E. Wolf and contributes to the thesis of Nhi Dao. It tackles a long-standing debate on SD of ACC and the role of PNS in water-born solid formation.

References: [1] Gebauer & Wolf, *JACS* **2019**, 141, 4490. [2] Gebauer et al., *Science* **2008**, 322, 1819. [3] Clark et al., *Sci. Rep.* **2022**, 12, 6870. [4] Duchstein et al., *Angew. Chemie IE* **2022**, 60, e202208475. [5] Higgins et al., *Macromolecules* **2020**, 53, 4137. [6] https://www.sasview.org/docs/user/models/spinodal.html and refs therein.